

MTH 253 – Unit #1 Written Homework

Evaluate the limits below using any technique we learned in this module. If you use L'Hospital's Rule, be sure to explicitly show how this rule applies to this limit.

#1.

$$\lim_{x \rightarrow 5} \frac{x^2 - 2x - 15}{x^2 - 25}$$

#2.

$$\lim_{x \rightarrow \infty} \frac{3x^5 + 9x^4 + x^3}{x^6 + 8x + 1}$$

#3.

$$\lim_{x \rightarrow \infty} \frac{e^{2x}}{\ln(x) + 2}$$

#4.

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

#5. Determine if the integral is convergent or divergent. If it is convergent, find the value.

$$\int_1^3 \frac{1}{(3-x)^2} dx$$